ENGR 103: Engineering Computation and Algorithmic Thinking

Homework Assignment #3 – Quality Control

Part 1 – Design

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Student's Name: _____

Step 1 – Understanding the Problem (7.5 pts)
Use the spaces provided below to answer the following questions:

- (1.5 pts) Do you understand everything in the problem? List anything you do not fully understand.

I understand everything in the problem.

- (2 pts) What are the functional requirements of the program, i.e., what does it need to do?

The function needs to take inputs for 5 measure of interest samples, the desired average of those samples, and the desired standard deviation of those sample. Then it calculates the actual average and standard deviation for the samples and compares, returning those values and informing the user if the measured value of the average is above, below, or equal to the desired average.

- (2 pts) What assumptions are you making?

We are assuming that the normal probability distribution is a good representative distribution for the measure of interest.

- (2 pts) What are the inputs, outputs, etc.?

Inputs:

1. Measure of interest 1
2. Measure of interest 2
3. Measure of interest 3
4. Measure of interest 4
5. Measure of interest 5
6. Desired average value
7. Desired standard deviation

Outputs:

- Actual average value
- Actual lower bound
- Actual upper bound
- Actual average value comparison to desired average value

8.

<u>Step 2 – Devise a Plan (10 pts)</u>

Use the space provided below to answer the following questions:

- (1.5 pts) What are the decisions that need to be made in this program?

How to gather input

How to ensure only numbers greater than 0 are input

How to calculate actual values

- (2 pts) What is the sequence of steps you need to complete?

Gather input

Ensure input is in the correct form

Calculate actual values

Output actual values and comparison to desired values

- (1.5 pts) How are you going to do the required calculations?

I will save the inputs as variables and then write out the equations to be performed with those variables.

(5 pts) Based on your answers to the questions in Step 2, provide an algorithm as **pseudocode** or as a **flowchart** showing the specific steps that are needed to create the required computer program. **Be very explicit!!!**

Input measure of interest 1

Input measure of interest 2

Input measure of interest 3

Input measure of interest 4

Input measure of interest 5

Input desired sample average value

Input desired standard deviation value

After each value is input make sure that it is greater than 0

Calculate sample average

Calculate lower bound

Calculate upper bound

Output actual lower bound

Output actual sample average

Output actual upper bound

Output actual sample average relation to range

<u>Step 4a – Looking Back (7.5 pts)</u>

Create a test plan (on paper) with test cases (i.e., good, bad, and edge cases).

- What are the good, bad, and edge test cases for ALL input in the program?
- What do you expect the results to be in each test case?

Use the table shown below to organize your proposed test cases. Test Case #1 is provided only as an example. You need to provide <u>at least five more test cases</u>. Note: Your test cases must incorporate the values of numerical inputs and numerical results your program will request and/or produce.

If you want to see more test case examples, consult the documents included in the module titled "POLYA 4SDP AND PYTHON STYLE GUIDELINES" in the course's Canvas website.

Test Case #	Value(s)	Expected	Good or Bad?
1	*n = 1,* *test score = 100*	*The average test score should be 100.*	*Good*
2	**n = 5** **test scores = 2.10, 2.12, 2.29, 2.15, 2.20** **desired mean = 2.2** **desired standard deviation = 0.10**	**Average = 2.066** **Lower bound = 2.066** **Upper bound = 2.334** **Within range**	**Good**
3	**n = 5** **test scores = 5.50, 5.60, 5.70, 5.80, 5.90** **desired mean = 5.00** **desired standard deviation = 0.20**	**Average = 5.700** **Lower bound = 4.731** **Upper bound = 5.269** **Above range**	**Good**
4	**n = 5** **test scores = 1.00, 1.01, 0.99, 1.02, 1.00** **desired mean = 2.00** **desired standard deviation = 0.15**	**Average = 1.004** **Lower bound = 1.799** **Upper bound = 2.201** **Below range**	**Good**

5	**n = 5** **test scores = 2.10, -1.00, 2.29, 2.15, 2.20** **desired mean = 2.20** **desired standard deviation = 0.10**	**Error – program** **ends/repeats asks for** **better input** **-1.00 !> 0**	**Bad**
6	**n = 5** **test scores = 3.134, 3.134, 3.134, 3.134, 3.134** **desired mean = 3.00** **desired standard deviation = 0.10**	**Average = 3.134** **Lower bound = 2.866** **Upper bound = 3.134** **Within range (boundary)**	**Edge**
7			
8			
9			